

# Odds and Ends & The Expenditure Function

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# The indirect Utility function and the meaning of $\lambda$ .

- Let  $x^M$  and  $y^M$  denote the optimal values (demands) in the utility maximization problem. and let  $\lambda^*$  denote the optimal value of the multiplier
- Notice that we can write

$$\begin{aligned} V(p_x, p_y, I) &= U(x^M, y^M) = L(x^M, y^M, \lambda^*) \\ &= U(x^M, y^M) + \lambda^*[I - p_x x^M - p_y y^M] \end{aligned}$$

- We want to calculate  $\frac{\partial V}{\partial I}$ , i.e. how utility changes with respect to changes in income.

$$\begin{aligned} \frac{\partial V}{\partial I} &= U_x \frac{\partial x^M}{\partial I} + U_y \frac{\partial y^M}{\partial I} + \lambda^* \left( 1 - p_x \frac{\partial x^M}{\partial I} - p_y \frac{\partial y^M}{\partial I} \right) + (I - p_x x^M - p_y y^M) \frac{\partial \lambda^*}{\partial I} \\ &= \underbrace{(U_x - \lambda^* p_x)}_{=0 \text{ from the FOC's}} \frac{\partial x^M}{\partial I} + \underbrace{(U_y - \lambda^* p_y)}_{=0 \text{ from the FOC's}} \frac{\partial y^M}{\partial I} + \underbrace{(I - p_x x^M - p_y y^M)}_{=0 \text{ from the FOC's}} \frac{\partial \lambda^*}{\partial I} + \lambda^* \\ &= \lambda^* \end{aligned}$$

- Thus,  $\lambda^*$  is the marginal value, in terms of utility of a marginal increase in income.
- The result we just proved is a version of the Constrained Envelope Theorem (NS Ch. 2 pp. 45–46)

# Odds and ends about the Lagrangean method

- Non-negativity constraints: Consider the cases we have seen that yield corner solutions, i.e. Perfect Substitutes or the preference from Homework 2, exercises 3,4.
- For these cases, it is not the case that the condition  $MRS = p_x/p_y$  holds.
- The reason is that consumption of one of the goods ends up being 0, i.e. at a **corner**.
- We can easily incorporate this case into our development of the lagrangian.
- Formally, we add **Non-negativity constraints**.

# Non-negativity constraints

- We have the following problem

$$\begin{aligned} \max_{(x,y) \in \mathbb{R}^2} : \quad & U(x, y) \\ \text{subject to} \quad & p_x x + p_y y \leq I \\ & x \geq 0 \\ & y \geq 0 \end{aligned}$$

- The Lagrangean is

$$L(x, y, \lambda, \lambda_x, \lambda_y) = U(x, y) + \lambda(I - p_x x - p_y y) + \lambda_x x + \lambda_y y$$

- First order conditions are:

$$\begin{aligned} \frac{\partial L}{\partial x} &= U_x(x^*, y^*) - \lambda^* p_x + \lambda_x^* = 0 \\ \frac{\partial L}{\partial y} &= U_y(x^*, y^*) - \lambda^* p_y + \lambda_y^* = 0 \end{aligned}$$

Plus the complementary slackness conditions

$$\lambda^* \geq 0, \lambda_x^* \geq 0, \lambda_y^* \geq 0, \quad \lambda^*[I - p_x x^* - p_y y^*] = 0, \quad \lambda_x^* x^* = 0, \quad \lambda_y^* y^* = 0$$

# The Dual of the Consumer Problem: Expenditure Minimization

- Any constrained maximization problems have associated dual constrained minimization problems.
- We have been considering until now the UMP.

$$\max_{x,y} U(x,y) \quad st. \quad p_x x + p_y y \leq I.$$

- The UMP also has an associated *dual* minimization problem which has an important economic meaning, the **Expenditure Minimization Problem**.
- As in any dual problem, the objective and the constraints are reversed.

$$\begin{aligned} \min_{(x,y) \in \mathbb{R}_+^2} : \quad & p_x x + p_y y \\ \text{subject to} \quad & U(x,y) \geq \bar{U} \end{aligned} \quad (\text{EMP})$$

where  $\bar{U}$  is a fixed level of utility.

# The Expenditure function

- Total expenditures are given by  $p_x x + p_y y$ .
- Thus, the minimized value of the expenditures, as a function of prices and the Utility level is

$$E(p_x, p_y, \bar{U}) = \min_{(x,y) \in \mathbb{R}_+^2} : p_x x + p_y y \quad (1)$$

subject to  $U(x, y) \geq \bar{U}$

- The interpretation,  $E(p_x, p_y, \bar{U})$  is the cheapest way in which you can achieve utility level  $\bar{u}$  at prices  $p_x, p_y$ .
- Armed with the expenditure function  $E$ , we can now answer the question: *How should we compensate in monetary terms an individual after a price change?*

# Relationship between Expenditure Function and Indirect Utility

- The expenditure function and the Indirect utility function are inverses of one another in the following sense: If  $\bar{U} = V(p_x, p_y, I)$  for some prices and income  $I$  then  $E(p_x, p_y, \bar{U}) = I$ , and if  $\bar{I} = E(p_x, p_y, U)$  for some prices and utility level  $U$  then  $U = V(p_x, p_y, \bar{I})$ .
- This relationship is very important because it allows us to analyze exactly how an individual responds to price changes.
- The relationship in the above bullet also allows us to obtain quite easily the expenditure function from the Indirect Utility Function quite easily.
- Let's see some examples.

## Some examples of Indirect Utility and Expenditure functions

- Cobb-Douglas case:  $U(x, y) = x^\alpha y^{1-\alpha}$ , the demands as we have seen are:

$$x^m(p_x, p_y, I) = \frac{\alpha I}{p_x}, \quad y^m(p_x, p_y, I) = \frac{(1-\alpha)I}{p_y}$$

- Thus the indirect utility function is:

$$V(p_x, p_y, I) = \frac{\alpha^\alpha (1-\alpha)^{1-\alpha} I}{p_x^\alpha p_y^{1-\alpha}}$$

- Using our result about expenditure and indirect utility functions we have that:

$$U = \frac{\alpha^\alpha (1-\alpha)^{1-\alpha} E(p_x, p_y, U)}{p_x^\alpha p_y^{1-\alpha}} \rightarrow E(p_x, p_y, U) = \frac{p_x^\alpha p_y^{1-\alpha} U}{\alpha^\alpha (1-\alpha)^{1-\alpha}}$$

# Perfect Complements

- $U(x, y) = \min \alpha x, \beta y, \alpha, \beta > 0$ .
- The demands, as we have seen are:

$$x^m(p_x, p_y, I) = \frac{\beta I}{\beta p_x + \alpha p_y}, \quad y^m(p_x, p_y, I) = \frac{\alpha I}{\beta p_x + \alpha p_y}$$

- Thus, the Indirect Utility function is:

$$V(p_x, p_y, I) = \min \left( \alpha \frac{\beta I}{\beta p_x + \alpha p_y}, \beta \frac{\alpha I}{\beta p_x + \alpha p_y} \right) = \frac{\alpha \beta I}{\beta p_x + \alpha p_y}$$

- Again, using our result about the relationship between expenditure and indirect utility we have

$$U = \frac{\alpha \beta E(p_x, p_y, U)}{\beta p_x + \alpha p_y} \rightarrow E(p_x, p_y, U) = \frac{\beta p_x + \alpha p_y}{\alpha \beta}$$

# Perfect Substitutes

- $U(x, y) = \alpha x + \beta y$ ,  $\alpha, \beta > 0$ .
- Remember that in this case we have corner solutions:
  - If  $\frac{\alpha}{\beta} > \frac{p_x}{p_y}$  then  $x^M(p_x, p_y, I) = I/p_x$ ,  $y^M(p_x, p_y, I) = 0$ .
  - If  $\frac{\alpha}{\beta} < \frac{p_x}{p_y}$  then  $x^M(p_x, p_y, I) = 0$ ,  $y^M(p_x, p_y, I) = I/p_y$ .
  - If  $\frac{\alpha}{\beta} = \frac{p_x}{p_y}$ ,  $x^M \in [0, I/p_x]$  and  $y = \frac{I - p_x x^M}{p_y}$ .
- Thus we have

$$V(p_x, p_y, I) = I \cdot \max\left(\frac{\alpha}{p_x}, \frac{\beta}{p_y}\right) = \frac{I}{\min\left(\frac{p_x}{\alpha}, \frac{p_y}{\beta}\right)}$$

- Thus we have that

$$E(p_x, p_y, U) = \min\left(\frac{p_x}{\alpha}, \frac{p_y}{\beta}\right) \cdot U$$

# Properties of Expenditure functions

- Homogeneity of degree 1: Intuitively, doubling all prices will precisely double expenditures needed to achieve a level of utility.
- Non-decreasing in prices: Intuitively, if one price has a positive increase, total expenditures must not go down.
- Concavity in prices: Although concavity is a technical property, this property can be explain intuitively graphically.